Recursion in Python :-

Recursion is a programming technique in which one function calls itself again & again & function call exists in the function definition.

eg. Normal function

```
def hello():
    print("hello")

hello()
```

fun def

fun call

o/p - "hello"

eg. Recursive function

```
def hello():
    print("hello")
    hello()
```

← function

← recu. call

```
hello()
```

Recursion Syntax :

fonction -

recursive call.

Base condition/ Stopping condition — Using return Keyword we stop the recursive call at particular condition.

```
def hello():
    print("hello")
    return "hello world"

result = hello()
```

```
def fact(n):
    if n == 1:
```

1 == 1

return 1

return n * fact(n-1)

multiplication recursive call.

num = 5

Print(fact(num))

5

5

4

fact(5-1)

return n * fact(n-1)

- fact(5)

return 5 * fact(4) = 120

fact(5) 5 * 24 = 120

- fact(4)

return 4 * fact(3)

fact(4) = 4 * 6 = 24

- fact(3)

return 3 * fact(2)

fact(3) = 3 * 2 = 6

- fact(2)

return 2 * fact(1)

fact(2) = 2 * 1 = 2

- fact(1)

return 1 = 1

fact1 = 1